



Phototransistors and Integrated Circuits

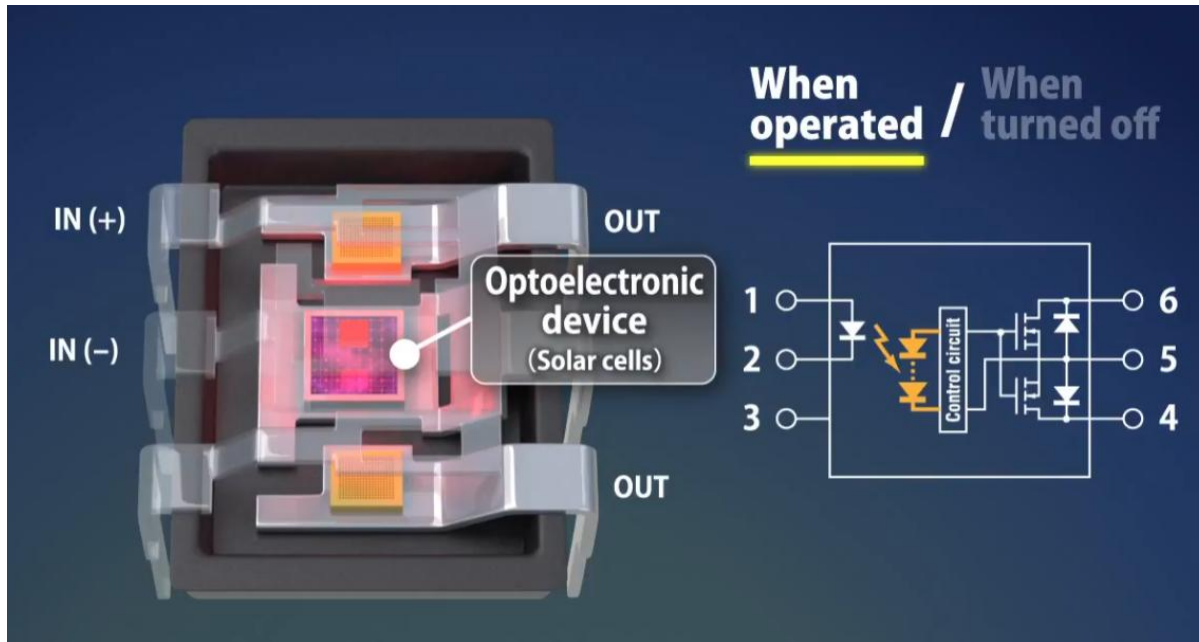
Spring 2026

Fundamental of Optoelectronics

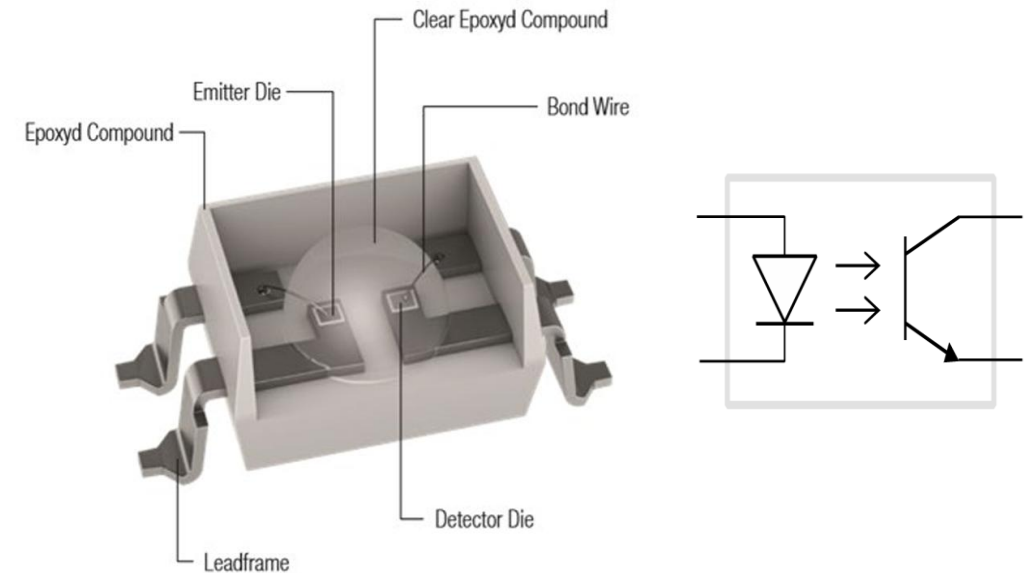
William Mattis

Photo Emitter & Receivers

- ❑ Photo-Relays are used in my senior design project to provide isolation against 600V and 12V Circuits
- ❑ Photodiodes and mosfet circuits as well as phototransistors can both be used to accomplish this effect



PhotoMOS semiconductor relays



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Background

□ Significance

- Backbone of Photonic Integrated Circuits and CMOS Integration
- Logical information can be sent at the speed of light
- No mechanical parts to wear out

□ History

LED:

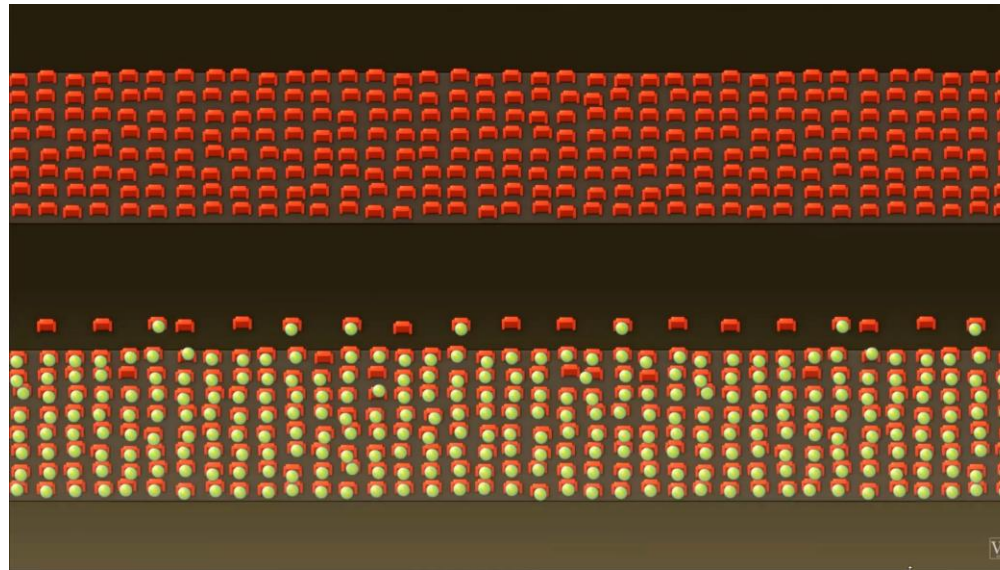
- Henry Joseph Round (1907) – First published report on LEDs
- Oleg Losev (1928) – First correct theory of LED operation
- Materials Breakthrough (1962) – GaAs LED construction at MIT

Photodetector:

- Elster and Geiter (1913) – Gaseous Ionization Detector
- RCA Labs (1936) – Photomultiplier Tube created voltage output
- Shockley Research Lab (1960s) – Solid state photon detectors

Basic Principles/Ideas

- Emission from LED:
 - Electrical charge is applied across LED, the current injects electrons into the conduction band and holes into the valance band. When electrons transition from the conduction band to the valance band, producing a packet of light with a wavelength determined buy the bandgap within the diode



Why It Was Almost Impossible to Make the Blue LED

Basic Principles/Ideas

- Phototransistor:
 - Light entering the phototransistor will reflect back and forth through the Semiconductor Film where the incoming light will induce a current through the “base” region connecting the collector to the emitter.
 - The light induces a generation process that increases the minority carrier concentration in the base region, creating electron-hole pairs according to the absorption coefficient.
 - A photocurrent acts as a base current with the finite photocarrier lifetime creating a photoconductive gain proportional to τ/τ_t .

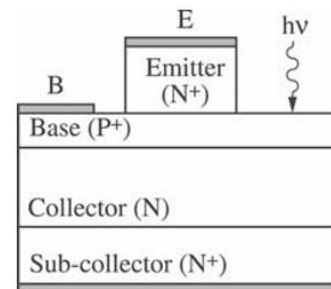
$$\text{gain} \sim \frac{\tau}{\tau_t}$$

$$I_C \approx \beta I_{\text{photo}}$$

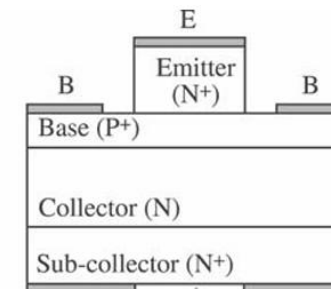
a) Front Facing

b) Rear Facing

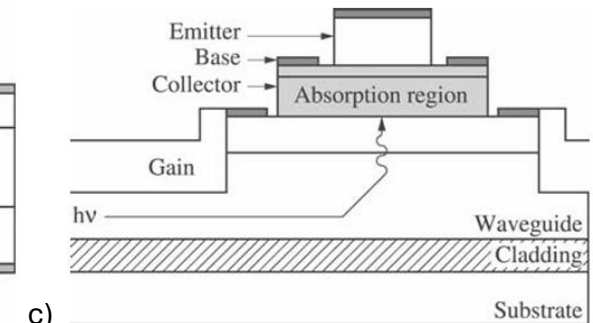
c) Lateral Illumination
with waveguide



a)



b)



c)

Derivation of Phototransistor Equations

Conductivity – Active Elements:

$$\sigma = qp^*\mu \quad \sigma = qp\mu^*$$

Carrier Density Effective Mobility

$$\Delta\sigma = q\mu^*\Delta p + qp\Delta\mu^*$$

Specific Responsivity (R_s):

$$R_s = \Delta\sigma / 4\sigma J \rightarrow R_s = \frac{(1+B)\eta\tau}{4pdh\nu [1 + (\omega\tau)^2]}$$

$$R_s \approx \frac{(1+B)\eta\tau}{4pdh\nu} \quad \omega \rightarrow 0$$

LED Photon Emission (Φ):

$$\Phi_{LED} = \eta_{LED} \frac{I_F}{q}$$

$$\Phi_{useful} = \Gamma_{oc} \eta_{path} \Phi_{LED}$$

Barrier Modulation (B):

$$B = \frac{\Delta\mu^*/\mu^*}{\Delta p/p} \rightarrow \Delta\sigma = q\mu^*(1+B)\Delta p$$

Change in Carrier Density (Δp):

$$\frac{d\Delta p}{dt} + \frac{\Delta p}{\tau} = g \quad \begin{array}{l} g - \text{generation rate} \\ \tau - \text{effective lifetime} \end{array}$$

$$g = \eta J / h\nu d \quad \Delta\sigma \approx q\mu^*(1+B) \frac{\eta J}{h\nu d} \tau$$

Steady State Conductivity

Apparent Quantum Efficiency (η^*):

$$\eta^* = \frac{\Delta I_{\text{electron current}}}{\Delta I_{\text{photon current}}} = \frac{\Delta\sigma dwl/q}{Jwl/h\nu}$$

$$\eta^* = \frac{(1+B)\eta}{\tau_t} \rightarrow \eta^* \propto (1+B)\eta \frac{\tau}{\tau_t}$$

Detector Generated Current (I_{det}):

$$I_{det} \propto q \Phi_{useful} \eta^*$$

$$I_{det} \propto q \left(\Gamma_{oc} \eta_{path} \eta_{LED} \frac{I_F}{q} \right) \eta^*$$

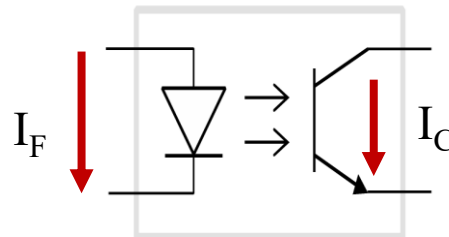
$$I_{det} \propto \Gamma_{oc} \eta_{path} \eta_{LED} (1+B) \eta \frac{\tau}{\tau_t} I_F$$

Current Transfer Ratio (CTR):

- Current gain of the optocoupler, equivalent to β in transistors

$$I_C \approx \beta I_{det,base}$$

$$CTR = \frac{I_C}{I_F}$$



$$CTR = \frac{I_C}{I_F} \propto \beta \Gamma_{oc} \eta_{path} \eta_{LED} (1+B) \eta \frac{\tau}{\tau_t}$$

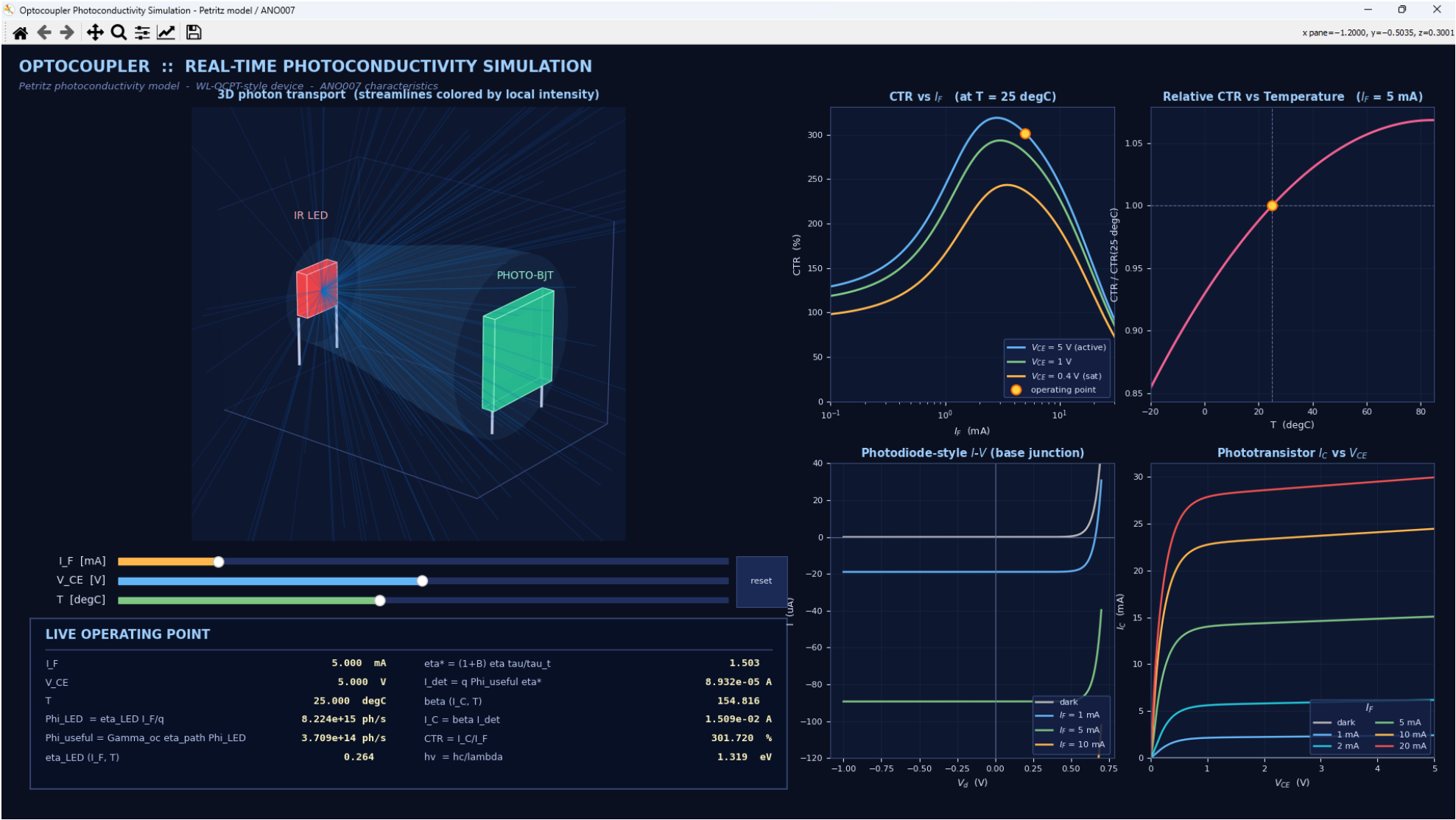
Phototransistor Operation

□ Material Differences

Material	Wavelength	Band Gap	Fiber Attenuation	Absorption Coefficient
Si	0.6 μm – 0.8 μm	1.12eV	--	--
GaAs	0.85 μm	1.43eV	2dB/km	$1 * 10^4 \text{ m}^{-1}$
InP	1.3 & 1.55 μm	0.75eV	0.5 & 0.2dB/km	$1.16*10^4$ & $6.8*10^3 \text{ m}^{-1}$

- InP:
 - 2 useable windows for optical fiber transmission
 - 1.3 μm : dispersion minimum
 - 1.55 μm : attenuation minimum
 - These are factors caused by the waveguide
- Operation Limiting factors:
 - High frequencies: >100MHz
 - Photo carriers in low field regions
 - Poor coupling between incident light & gate junction

Animation



Application

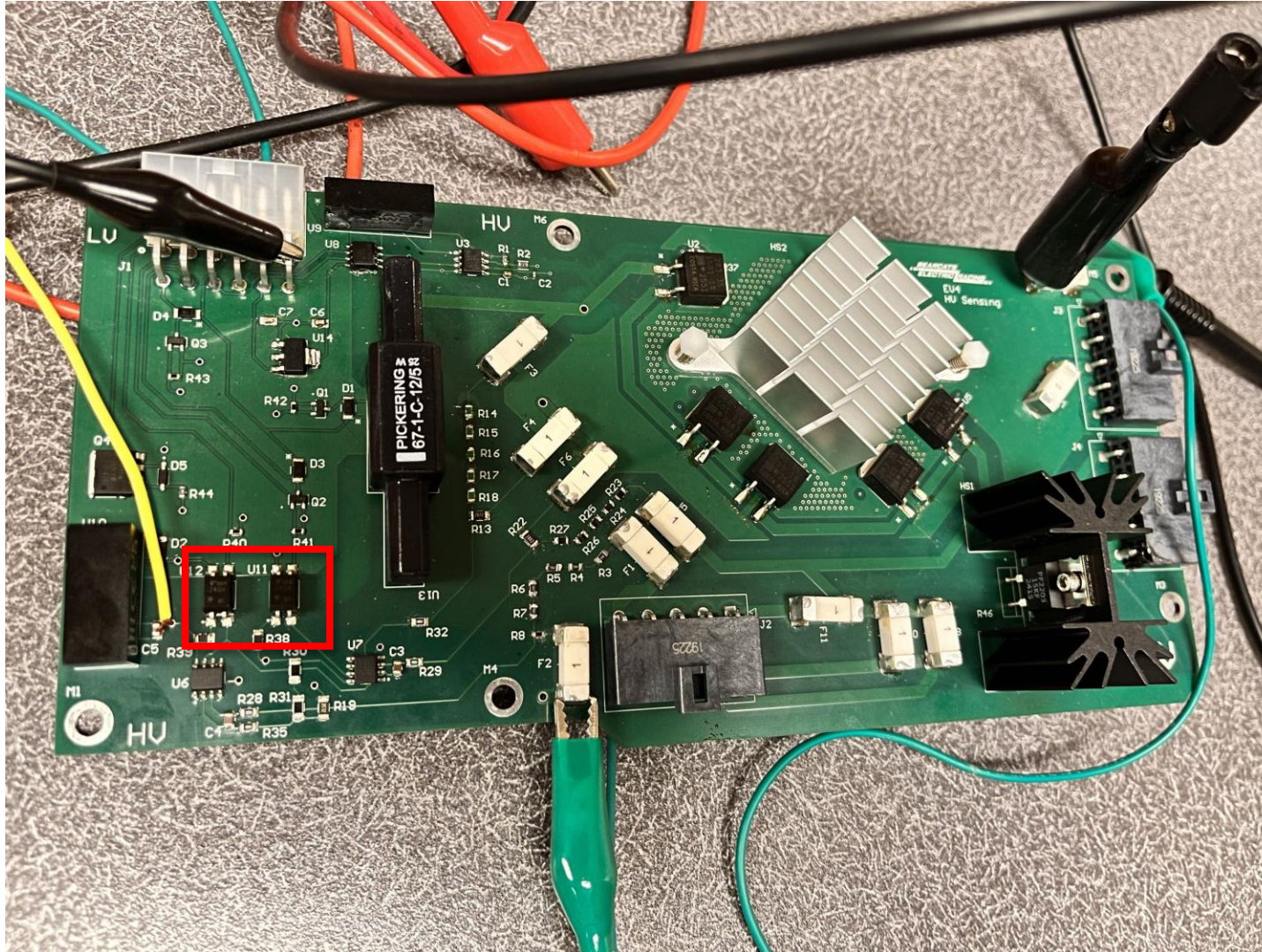


Figure 29: Turn-off example (WL-OCPT 817 Bin B) (V_{RF} , V_{RCI})

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Practical Operation:
[Project HV Sensing board-EV4 - Fixed](#)

Annotations

Name of article	Link
Numerical Modeling of Silicon Photodiodes for High Accuracy Applications with PC-ID	https://nvlpubs.nist.gov/nistpubs/Legacy/IR/nistir4592.pdf
ANO007 Understanding Phototransistor Optocouplers	https://www.welcome.com/components/media/o760909v410%20ANO007a_EN.pdf?srsltid=AfmBOoobbLzQ7Oepr1r0VjAEMVE3Nmj_T89iFHcQ0qi8LqT-0EA5M2CR
Theory of Photoconductivity in Semiconductor Films	https://journals.aps.org/pr/pdf/10.1103/PhysRev.104.1508
Phototransistors - Optoelectronic Sensors	https://lab.semi.ac.cn/library/upload/files/2021/10/26164535197.pdf?utm_source=chatgpt.com
Modeling of the temperature-dependent spectral response of In _{1-x} Ga _x Sb infrared photodetectors	https://ntrs.nasa.gov/api/citations/20080014302/downloads/20080014302.pdf
Why It Was Almost Impossible to Make the Blue LED	https://www.youtube.com/watch?v=AF8d72mA41M
PhotoMOS semiconductor relays	https://www.youtube.com/watch?v=7lhZ96ga-D8